

Big Data Analytics - Unit Wise Question Bank

Unit 1: Introduction to Big Data

1. Define Big Data and explain its characteristics: Volume, Velocity, Variety, Veracity, and Value.
2. Differentiate between structured, semi-structured, and unstructured data with examples.
3. Explain the evolution of Big Data.
4. Discuss the challenges associated with Big Data.
5. Why is Big Data important in modern industries? Explain with examples.
6. Compare Traditional Business Intelligence (BI) and Big Data Analytics.
7. Explain industry applications of Big Data.
8. What is Data Science? Explain its relationship with Big Data.
9. Write short notes on Big Data Analytics.
10. Explain the role of Big Data in decision making.

Unit 2: NoSQL Data Management

1. What is NoSQL? Explain its features and need.
2. Explain different types of NoSQL databases.
3. Compare SQL, NoSQL, and NewSQL databases.
4. Explain key-value data model with examples.
5. Discuss document-based databases.
6. Explain graph databases and their applications.
7. What are aggregates in NoSQL databases?
8. Explain map-reduce in NoSQL systems.
9. Describe partitioning and combining techniques in NoSQL.
10. List advantages and limitations of NoSQL databases.

Unit 3: Basics of Hadoop

1. Explain Hadoop and its history.
2. Why Hadoop is required for Big Data processing?
3. Compare RDBMS and Hadoop.

4. Explain Hadoop architecture with diagram.
5. Describe the components of Hadoop ecosystem.
6. Explain HDFS architecture and working.
7. What are the key features and advantages of Hadoop?
8. Explain data formats supported in Hadoop.
9. Describe processing of data with Hadoop.
10. Explain MapReduce workflow.
11. Describe anatomy of a MapReduce job run.
12. Explain shuffle and sort phase in MapReduce.
13. Discuss task execution in MapReduce.
14. Explain input and output formats in Hadoop.

Unit 4: Hadoop Related Tools

1. Explain the architecture and features of HBase.
2. What is Pig? Explain Pig data model.
3. Discuss Pig Latin with examples.
4. Compare Pig and Hive.
5. Explain Hive architecture and components.
6. Describe Hive data types and file formats.
7. Explain HiveQL data definition commands.
8. Discuss HiveQL data manipulation commands.
9. Write HiveQL queries for database operations.
10. Explain the use of JSON in Hadoop ecosystem.
11. Discuss Cassandra and its features.
12. What are Jasper Reports? Explain their applications.